

Resume

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PROFESSIONAL SUMMARY

Mechatronics Engineer with 9+ years of experience specializing in robotic cell automation, PLC programming (TIA Portal, Beckhoff), and plant commissioning. Proven track record in reducing cycle times and integrating advanced vision systems. Seeking to leverage technical expertise in robotics and Industry 4.0 to drive operational efficiency.

EDUCATIONAL QUALIFICATIONS

Degree / Certificate	Institution/ Board	Year
1. M.Tech (Digital Manufacturing)	Birla Institute of Technology And Science - Pilani	Pursuing
2. B.Tech (Mechatronics)	Lovely Professional University Phagwara	2017
3. SSC (CBSE)	BCM Senior Secondary School, Ludhiana	2013
4. HSC (PSEB)	RS Model Senior Secondary School, Ludhiana	2011

TECHNICAL SKILLS

1. Robot teaching and programming having experience of ABB - KUKA- Fanuc programming.
2. Experienced in robotic and automated plant troubleshooting.
3. Integration of Python with Siemens PLC.
4. Integration of vision systems, including object detection.
5. Safety module implementation.
6. PLC programming and logic forming capabilities having proficiency in TIA Portal, Beckhoff
7. Electrical and electronic breakdown handling capabilities with experience in foundry, FMCG packing line, Robotic lines.
8. Foundry technology with hands-on experience of Loramendi Coldbox foundry process.
9. Cycle time study and controlling.
10. Familiarity of HMI/Scada , TIA Portal, ABB Robot Studio, Automated machine handling .
11. Basic understanding of proportional valves, pressure transmitters, encoders, Linear transducers.
12. Basic understanding and hands-on experience of Foreign machines.
13. Inventory and manpower handling capabilities.
14. Production planning and control skills.
15. Basic knowledge for development of new products for robotic systems.
16. Creative thinking and always ready to explore new challenges.

Machine/Tool handling: Robot Teaching, TIA Portal, Beckhoff ,
Loramendi Foundry process, Welding, Casting,, Lathe, Milling Machine, Arduino
Software: Robot Studio, Work Visual TIA Portal, TwinCAT, MATLAB, Solid Works, Proteus, Atmel Studio
Languages: C, C++

Work Experience

Designation: Associate Manager at **Godrej & Boyce Mfg. Co. Ltd**

Duration: January, 2026 to Present

Description: Continuing to handle the project team for Infrastructure, capacity and capability development, Automation, Robotics and Plant Commissioning.

Designation: Deputy Manager - Projects & Automation at **Godrej & Boyce Mfg. Co. Ltd**

Duration: June, 2022 to December, 2025

Description: Working as Deputy Manager for Interio division of G&B where I am handling the project team for Infrastructure, capacity and capability development, Automation, Robotics and Plant Commissioning.

Achievements:

1. Project Commissioning for Cupboard assembly line for Khalapur Plant.
2. Robotic chemical Foam pouring setup for Haridwar Plant.
3. Robotic Releasing agent pouring for foaming line.
4. SVG system for Haridwar Plant.
5. Implementation Vision system for dynamic robot movement programming.
6. Automatic Shutter bending and corner forming
7. In-house development of vision system for automatic FG transfer.

Designation: Plant Engineer at **Bharat Milk Products Pvt Ltd**, Kotkapura

Duration: October, 2019 to June, 2022

Description: Was leading maintenance department in Bharat Milk Products Pvt Ltd where I was responsible for maintenance and troubleshooting of automatic dairy technology machines, utility department and ETP plant.

Achievements:

1. 2 times appreciation award for good management practices.
2. CIP plant commissioning and installation with successful trials.
3. Calibration of CTQ components for the assurance of quality.
4. Formation of SOPs for plant activities.
5. Formation of ERP program for handling plant activities online.

Designation: Mechatronics Engineer at **DCM Engineering Products**, Ropar

Duration: October, 2017 to June, 2019.

Description: Worked as Mechatronics Engineer in DCM where I am handling Loramendi Automatic Core making machines and ABB foundry plus robots. My main responsibilities were production control, inventory & manpower handling and general maintenance & troubleshooting of robotic equipment.

Achievements:

1. Teaching and programming of robots for Mahindra S101 Block Water-jacket and slab assembly.
2. Cycle time reduction of crank cell by 30%.
3. Achieved targeted cycle of 200 cycle per shift which was previously 120.
4. Process standardization by implementing QCS and WSS for Mahindra U108 and Eicher 110p block core making.

PROJECTS UNDERTAKEN

Project: CIP Plant commissioning and installation

Objective: To implement CIP plant as per requirement.

Description: Implemented CIP plant in Bharat Milk Products Pvt Ltd for automatic cleaning in place operation of whole plant through one equipment. Automated the whole process using Siemens Simatic S7-1200 PLC.

Project: Intelligent Pneumatic Braking System On Table Saw

Objective: To design a safety braking on Table Saw

Description: Designed and made a safety system for table saw using simple relay logic circuit, IR based sensor and pneumatic actuator in order to make a system that can prevent any casualties to the worker operating a table saw.

Project: 'Quad Copter'

Objective: To design the quad copter using 2.4ghz flight controller

Description: Designed and made a quad copter using a 2.4ghz flight controller having range of about 30 meters through which it can hover easily and it can also perform stunts and flips.

Project : Line Follower Robot Using Atmega8 IC

Objective: To design a line follower robot

Description: Designed and made a line follower robot using Atmega8 programmable IC that can follow black line marked on white surface by using the logic circuits and optical sensors

WORKSHOPS AND CERTIFICATIONS

- Certified on **PLC**(Programmable Logic Controllers) from **TATA Steel**, 2020
- Certified on **Industry 4.0** from **TATA Steel**, 2020
- Certified on **TQM**(Total Quality Management) Principles from **TATA Steel**, 2020
- Trained on IRC 5 ABB robotware, **ABB**, 2018
- Workshop on 'Swarn Robotics', **Lovely Professional University**, 2014
- Workshop on 'Robotics workshop' from **Intel Certification**, Delhi, 2014

EXTRA CURRICULAR ACTIVITIES

Participated in:

- 5 Times Marathon Runner(5KM & 10KM) for Health and Anti-drug campaign.

- Debate, 'One India', Lovely Professional University,2014
- Duet Bhangra, 'Youth Vibe', Lovely Professional University,2014

AWARDS AND HONOURS

- Best in-house Automation By Godrej Interio, 2024
- Second Prize-Indo-US Robo League by Technophilia,2014
- First Prize - Science Fair By Punjab School Education Board,Punjab,2007,2008
- Third Prize – Green Olympiad,Ludhiana,Punjab,2008

LANGUAGES KNOWN:

English,Hindi,Punjabi,

HOBBIES:

Workout,Biking,Running

DATE : 14/08/2026

PLACE Ludhiana
Punjab

Amandeep Singh